

$$1 \quad m(X) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$m(a+bX) = a + b \cdot m(X)$$

$$m(a+bX) = a + b \left(\frac{1}{N} \sum_{i=1}^N x_i \right)$$

$$m(a+bX) = \left(\frac{1}{N} \sum_{i=1}^N a \right) + b \left(\frac{1}{N} \sum_{i=1}^N x_i \right)$$

$$m(a+bX) = a + bx_i$$

$$m(a+bX) = a + b(m(X))$$

$$m(a+bX) = m(a+bX)$$

$$2 \quad \text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y)) \rightarrow \text{cov}(X, a+bY) = b \cdot \text{cov}(X, Y)$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i - m(a+bY)) = b \cdot \text{cov}(X, Y)$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i) - (a + b \cdot m(Y)) = b \cdot \text{cov}(X, Y)$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(b(y_i - m(Y))) = b \cdot \text{cov}(X, Y)$$

$$b \left(\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y)) \right) = b \cdot \text{cov}(X, Y)$$

$$b \cdot \text{cov}(X, Y) = b \cdot \text{cov}(X, Y)$$

$$3 \quad s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2 \Rightarrow \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$$

$$\frac{1}{N} \sum_{i=1}^N ((a+bx_i) - m(a+bx))((a+bx_i) - m(a+bx)) = b^2 \text{cov}(X, X)$$

$$\frac{1}{N} \sum_{i=1}^N ((a+bx_i) - (a+b \cdot m(x)))((a+bx_i) - (a+b \cdot m(x))) = b^2 \text{cov}(X, X)$$

$$\frac{1}{N} \sum_{i=1}^N (b(x_i - m(x)))(b(x_i - m(x))) = b^2 \cdot \text{cov}(X, X)$$

$$b^2 \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(x))(x_i - m(x)) = b^2 \cdot \text{cov}(X, X)$$

$$b^2 \cdot \text{cov}(X, X) = b^2 \cdot \text{cov}(X, X)$$

$$\hookrightarrow \text{cov}(X, X) = s^2$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(x))(x_i - m(x)) = s^2$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2 = s^2$$

4 $m(g(X)) = g(m(x))$ is not always true because the mean of a transformed variable is not the same thing as a transformation of the mean. This would only be true if the transformation was linear.

Counter example: let $X = \{0, 4\}$; $g(x) = x^2$

1) Calculate $g(m(x))$:

$$m(X) = (0+4)/2 = 2$$

$$g(m(x)) = g(2) = 2^2 = 4$$

2) Calculate $m(g(x))$:

$$g(X) = \{0^2, 4^2\} = \{0, 16\}$$

$$m(g(x)) = (0+16)/2 = 8$$

Since $8 \neq 4$, $m(g(x)) \neq g(m(x))$ in general.